
From Click to Checkout : E-commerce Customer Journey Analysis

An Business analytics project analyzing how customers move from their first website visit to completing a purchase.

Project Overview

Every online purchase begins with a single click—but not every visitor becomes a customer. Understanding where customers arrive, how they navigate the website, which products capture their attention, and where they abandon their journey is essential for improving conversions and driving business growth.

This project analyzes the complete customer journey for Fuzzy Factory, combining Python (Pandas) for data preparation, MySQL for business analysis, and Power BI for interactive dashboards. The objective is to identify opportunities to improve customer engagement, increase conversion rates, and optimize business performance.

Business Objective

This analysis aims to answer the following business questions:

- Which marketing channels attract the highest-quality customers?
- How effectively does the website convert visitors into buyers?
- Which landing pages encourage customers to make a purchase?
- Which products generate interest but fail to convert into sales?
- How many customers return to purchase again?
- What opportunities exist to improve the customer journey from the first click to checkout?

Data Preparation

The raw datasets were imported into Python (Pandas), where missing values and data types were cleaned and standardized. The cleaned data was then loaded into a MySQL relational database, where primary and foreign key relationships were established to maintain data integrity.

The MySQL database was connected to Power BI, enabling the creation of interactive dashboards to analyze website traffic, customer behavior, product performance, sales trends, and conversion metrics.

Customer Journey Analysis

1. Acquiring Customers – Where Does the Journey Begin?

Every customer starts by visiting the website through a marketing channel. The first step was to evaluate which traffic sources generated not only visitors but actual customers.

While Google Search drives higher overall traffic and revenue, Bing Search achieves a significantly better conversion rate.

Customer Journey Analysis

2. Turning Visitors into Customers

After arriving on the website, the next objective is understanding how many visitors actually complete a purchase.

- 93% of Maven toy store sessions end with no purchase. Only 6% is the conversion rate.
- 52% users abandon the session before cart.

Customer Journey Analysis

3. The First Impression – Landing Page Performance

Landing pages create the customer's first impression and often determine whether visitors continue exploring or leave the website.

While /lander-5 generates the highest conversion rate, both Google Search and Bing Search route the majority of their sessions to /lander-2, followed by /home

Customer Journey Analysis

4. Product Discovery – Are Customers Finding the Right Products?

After entering the website, customers begin exploring products before making a purchase decision.

- The Original Mr. Fuzzy drives 60% of sales but has the lowest conversion rate (14.68%).
- The Hudson River Mini Bear has the highest conversion rate (22.26%) and lowest refunds (1.28%), but lacks traffic exposure.
- The Birthday Sugar Panda suffers from a store-high 6.04% refund rate, eroding profits.

Customer Journey Analysis

5. Building Customer Loyalty

Acquiring customers is only one part of sustainable growth. Long-term profitability depends on encouraging customers to return.

- 98% of the users never came back

Strategic Recommendations:

Six prioritized actions address the highest-volume losses, from channel spend to post-purchase retention.

- Shift channel spend: weigh budget toward Bing and direct (7%+ CVR);

Primary Segment: 26-48 Age Group

- ~~Fix landing pages: redirect to~~ Largest share of customer base with 55% representation (10.17%).
- Demonstrates highest transaction activity and product engagement
- Re-balance merchandising: reduce reliance on Mr. Fuzzy (60% of orders); grow high-converting niche bears

Strategic Recommendations:

- Reduce checkout friction: close the 52% product-to-cart drop and 37% billing abandonment
- Address refunds: investigate Birthday Primary Segment: 26-48 Age Group
Sugar Panda quality issues at a 6.04% refund rate
- Largest share of customer base with 55.5% representation
- Build retention: target 31,105 one-time buyers via post-purchase email; repeat base is only 591